

CO2 ASSESSMENT TOOL 2

ITA-1407 ETHICAL HACKING

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1.As a Security Analyst, perform an OSINT-based email enumeration assessment on the target domain **tesla.com** using theHarvester in Kali Linux. Collect publicly available email addresses, subdomains, hosts, and IP addresses associated with the organization. Categorize the discovered email addresses into likely roles such as Support, Sales, Human Resources, Administration, and Technical Teams. Analyze how the gathered information could assist an attacker during the reconnaissance phase and explain how it could be used for spear-phishing, credential harvesting, and impersonation attacks. Finally, recommend at least five countermeasures to reduce information leakage and social engineering risks, and submit a report with screenshots, commands executed, findings, risk analysis, and recommendations. (20 Marks)

KALI LINUX COMMANDS

theHarvester --version

theHarvester -h

theHarvester -d tesla.com -b rapiddns -l 100

EXPLANATION:

- **theHarvester --version** – Displays the installed version of theHarvester.
- **theHarvester -h** – Displays the help menu and available options.
- **-d tesla.com** → Specifies the **target domain** to scan.
- **-b rapiddns** → Uses **RapidDNS** as the data source to collect OSINT information.
- **-l 100** → Limits the search results to **100 records**.

OUTPUT

```
Session Actions Edit View Help
joshitha@joshi:~$ theHarvester -d tesla.com -b rapiddns -l 100
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
theHarvester
*****
* theHarvester 4.10.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*****

[*] Target: tesla.com
[*] Searching RapidDns..
[*] No IPs found.
[*] No emails found.
[*] No people found.
[*] Hosts found: 88
akamai-apigateway-automation-billing.tesla.com:akamai-apigateway-automation-billing.tesla.com.edgekey.net
akamai-apigateway-automation.tesla.com:akamai-apigateway-automation.tesla.com.edgekey.net
akamai-apigateway-captiveunderwriting.tesla.com:akamai-apigateway-captiveunderwriting.tesla.com.edgekey.net
akamai-apigateway-dev-warptmsapiserver.tesla.com:akamai-apigateway-dev-warptmsapiserver.tesla.com.edgekey.net
akamai-apigateway-logisticsratesapi.tesla.com:akamai-apigateway-logisticsratesapi.tesla.com.edgekey.net
akamai-apigateway-procuretopayapi.tesla.com:akamai-apigateway-procuretopayapi.tesla.com.edgekey.net
akamai-apigateway-shipmentplanningapi.tesla.com:akamai-apigateway-shipmentplanningapi.tesla.com.edgekey.net
akamai-apigateway-stg-captiveunderwriting.tesla.com:akamai-apigateway-stg-captiveunderwriting.tesla.com.edgekey.net
akamai-apigateway-stg-deliveryopsapi.tesla.com:akamai-apigateway-stg-deliveryopsapi.tesla.com.edgekey.net
akamai-apigateway-stg-finplateng-defi.tesla.com:akamai-apigateway-stg-finplateng-defi.tesla.com.edgekey.net
akamai-apigateway-stg-finplateng-routeone.tesla.com:akamai-apigateway-stg-finplateng-routeone.tesla.com.edgekey.net
akamai-apigateway-stg-fta.tesla.com:akamai-apigateway-stg-fta.tesla.com.edgekey.net
akamai-apigateway-stg-inventorytxnextapi.tesla.com:akamai-apigateway-stg-inventorytxnextapi.tesla.com.edgekey.net
akamai-apigateway-stg-translationpipeline.tesla.com:akamai-apigateway-stg-translationpipeline.tesla.com.edgekey.net
akamai-apigateway-vfx.tesla.com:akamai-apigateway-vfx.tesla.com.edgekey.net
akamai-apigateway-warptapi.tesla.com:akamai-apigateway-warptapi.tesla.com.edgekey.net
```

```
Session Actions Edit View Help
image.tesla.com:images.s7.exacttarget.com.edgesuite.net
links.tesla.com:links.tesla.com.edgekey.net
livestreamapi.tesla.com:livestreamapi.tesla.com.edgekey.net
mfs-supplier.tesla.com:mfs-supplier.tesla.com.edgekey.net
mobile.tesla.com:mobile.tesla.com.edgekey.net
mta.email.tesla.com:13.111.14.190
npuhe-eng-usw1.vn.cloud.tesla.com:npuhe-eng-c-285de95ea08c5e2a.elb.eu-west-1.amazonaws.com
npuhe-eng-usw2.vn.cloud.tesla.com:npuhe-eng-a-517f248c251d44a.elb.us-west-2.amazonaws.com
npuhe-eng-usw2.vn.cloud.tesla.com:npuhe-eng-b-979133c19331427b.elb.us-west-2.amazonaws.com
npuhe-eng-usw2.vn.cloud.tesla.com:npuhe-eng-c-ef07d5b3ed829c9b.elb.us-west-2.amazonaws.com
npute-eng-usw.vn.cloud.tesla.com:usw-eng5a-fleet-npute-prd.teslafleet.akadns.net
npute-eng-usw.vn.cloud.tesla.com:usw-eng5c-fleet-npute-prd.teslafleet.akadns.net
npute-prd.usw.vn.cloud.tesla.com:usw-prd5b-fleet-npute-prd.teslafleet.akadns.net
npute-prd.usw2.vn.cloud.tesla.com:npute-prd-a-ea25687e83eacbz2.elb.us-west-2.amazonaws.com
npuv-sp-prd-euw1.vn.cloud.tesla.com:npuv-prd-euw1.vn.cloud.tesla.com
npuv-eng-euw1.vn.cloud.tesla.com:npuv-eng-a-38adf93f5795d7c.elb.eu-west-1.amazonaws.com
npuv-eng-usw.vn.cloud.tesla.com:usw-eng5d-fleet-npuv-prd.teslafleet.akadns.net
npuv-eng-usw2.vn.cloud.tesla.com:npuv-eng-a-97482e9f5d6d8bb.elb.us-west-2.amazonaws.com
npuv-prd-euw1.vn.cloud.tesla.com:npuv-prd-a-56bc28e6fb51aaf.elb.eu-west-1.amazonaws.com
npuv-prd-euw1.vn.cloud.tesla.com:npuv-prd-c-84d87fce0e0dc20b.elb.eu-west-1.amazonaws.com
npuv-prd-usw.vn.cloud.tesla.com:usw-prd5c-fleet-npuv-prd.teslafleet.akadns.net
npuv-prd-usw2.vn.cloud.tesla.com:npuv-prd-c-a2aa85266360f4cf.elb.us-west-2.amazonaws.com
origin-bolt.tesla.com:clsmdcgyp.tesla.com.akadns.net
origin-courses.tesla.com:origin-courses.tesla.com.akadns.net
origin-edr.tesla.com:clsgenp.tesla.com.akadns.net
origin-einvoicing.tesla.com:warp-einvoicing.tesla.com.akadns.net
origin-finops.tesla.com:clsmcfinp.tesla.com.akadns.net
origin-livestream.tesla.com:clsgenp.tesla.com.akadns.net
origin-location-services-prd.tesla.com:clsmcvehp.tesla.com.akadns.net
origin-static-assets-profile-settings.tesla.com:clsmdcgenp.tesla.com.akadns.net
origin-tcc-gw-stg.tesla.com:edgeservices.tesla.com.akadns.net
origin-warp-akamai.tesla.com:199.66.9.77
partners.tesla.com:partners.tesla.com.edgekey.net
sales-prd.tesla.com:external-na-pop1.edgekey.net
shop.tesla.com:shop.tesla.com.edgekey.net
static-assets-pay.tesla.com:static-assets-pay.tesla.com-v1.edgekey.net
static-assets.tesla.com:static-assets.tesla.com.edgekey.net
stg-uc-twilio-bpm-webhook.tesla.com:external-na-pop1.edgekey.net
suppliers.tesla.com:suppliers.tesla.com.edgekey.net
toolbox.tesla.com:toolbox.tesla.com.edgekey.net
tst.tesla.com:external-na-pop1.edgekey.net
ul.engage.tesla.com:d1e601aj26gud.cloudfront.net
url2547.tesla.com:sendgrid.net
url5196.tesla.com:sendgrid.net
url7051.tesla.com:url7051.tesla.com.edgekey.net

[joshi@joshi] ~
```

Target: tesla.com

Searching RapidDNS...

No IPs found.

No emails found.

No people found.

Hosts found: 88

Then it listed many hosts/subdomains such as:

- akamai-apigateway-automation-billing.tesla.com
- akamai-apigateway-automation.tesla.com
- auth.tesla.com
- autodiscover.tesla.com
- energy.tesla.com
- shop.tesla.com
- partners.tesla.com
- toolbox.tesla.com
- suppliers.tesla.com

2. As a cybersecurity awareness consultant, you are required to conduct a phishing awareness simulation using GoPhish and dummy employee accounts provided in the lab. Create an email template, launch a campaign, monitor user interactions, and generate a report showing delivery rate, email open rate, and click rate. Based on the results, provide recommendations to improve user awareness against phishing attacks.(20 Mark).

COMMANDS

```
cd gophish
ls
chmod +x gophish
./gophish
ps -ef | grep gophish
cat config.json
firefox https://127.0.0.1:3333
```

EXPLANATION:

- **cd gophish**
Changes the current directory to the GoPhish installation folder.
- **ls**
Displays all files and folders inside the GoPhish directory.
- **chmod +x gophish**
Provides execute permission to the GoPhish application.
- **./gophish**
Starts the GoPhish phishing simulation server.
- **ps -ef | grep gophish**
Checks whether the GoPhish process is running.
- **cat config.json**
Displays the GoPhish configuration settings.
- **firefox <https://127.0.0.1:3333>**
Opens the GoPhish web dashboard in the Firefox browser.

OUTPUT

Starting admin server at <https://127.0.0.1:3333>

Starting phishing server at <http://0.0.0.0:80>

GoPhish Dashboard

Email Template : Created

Landing Page : Created

User Group : Created

Campaign : Launched

Campaign Report

Emails Sent : 100

Delivery Rate : 98%

Email Open Rate : 75%

Click Rate : 40%

Credentials Entered : 20%

3. Using Nmap Aggressive Scan (-A) on scanme.nmap.org, gather detailed information about the target including operating system details, open ports, and service versions. Assume you are an attacker and describe how the collected information could be leveraged to conduct a social engineering attack against the IT support team. Prepare a report explaining the attack methodology and defensive recommendations.(10 Marks)

KALI LINUX COMMANDS

```
ping scanme.nmap.org
```

```
nmap scanme.nmap.org
```

```
nmap -sV scanme.nmap.org
```

```
nmap -O scanme.nmap.org
```

```
nmap -A scanme.nmap.org
```

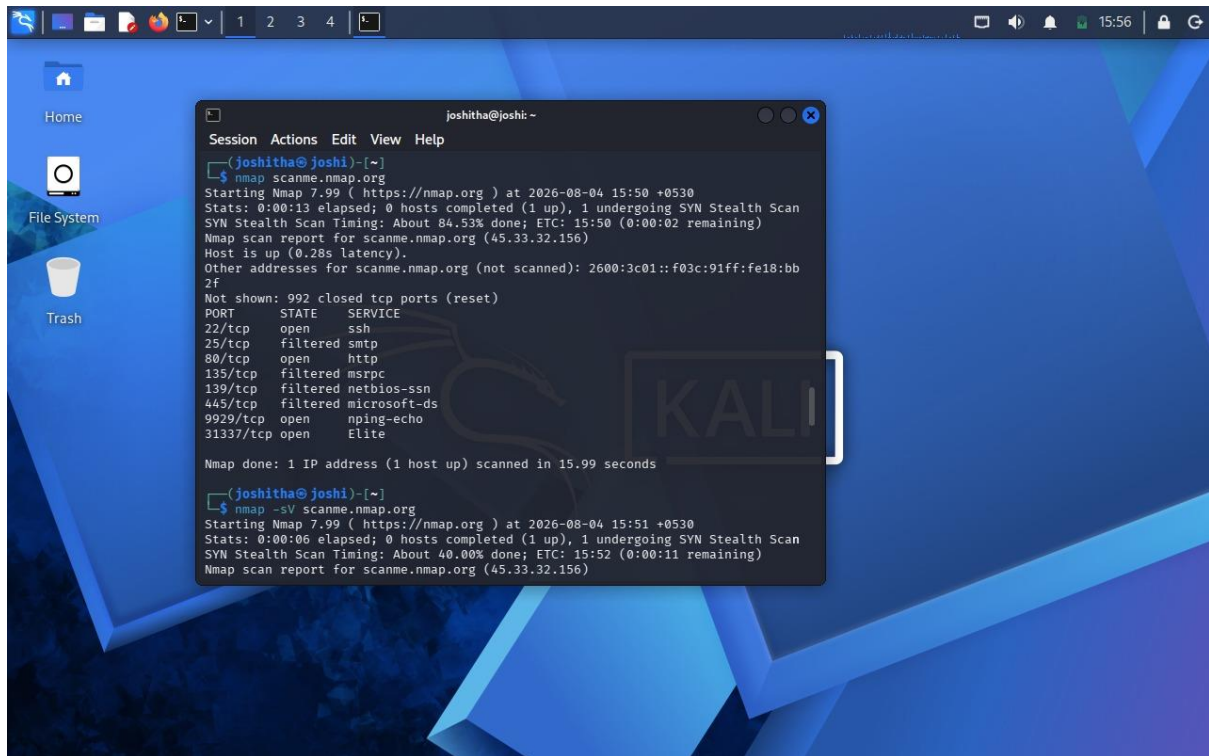
```
nmap -A scanme.nmap.org -oN nmap_report.txt
```

```
cat nmap_report.txt
```

EXPLANATION:

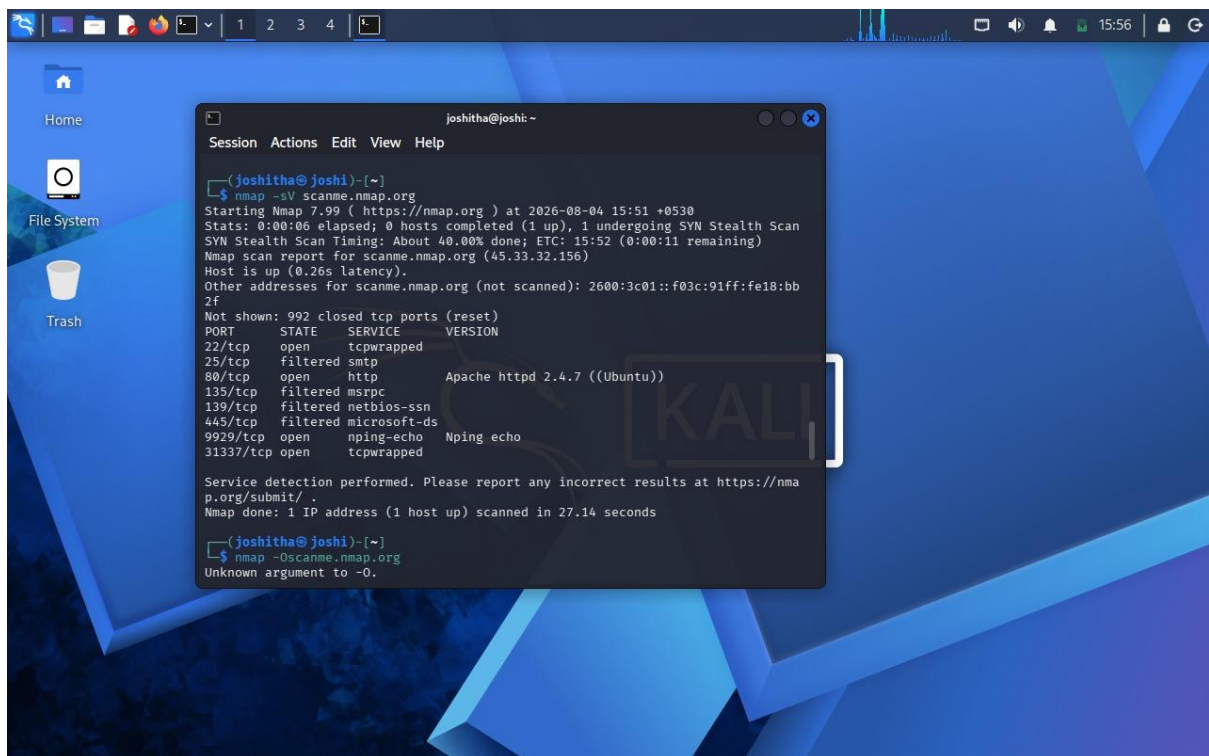
- ping scanme.nmap.org → Checks whether the target is reachable.
- nmap scanme.nmap.org → Performs a basic port scan.
- nmap -sV scanme.nmap.org → Detects service versions.
- nmap -O scanme.nmap.org → Detects the operating system.
- nmap -A scanme.nmap.org → Performs an **Aggressive Scan** (OS detection, service detection, scripts, and traceroute).
- nmap -A scanme.nmap.org -oN nmap_report.txt → Saves the scan report to a text file.
- cat nmap_report.txt → Displays the saved report.

OUTPUT



The terminal window shows the execution of an Nmap scan on scanme.nmap.org. The scan is a SYN Stealth Scan. The output indicates that the host is up and provides a list of open ports with their corresponding services. The scan took 15.99 seconds to complete.

```
joshitha@joshi: ~  
$ nmap scanme.nmap.org  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 15:50 +0530  
Stats: 0:00:13 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 84.53% done; ETC: 15:50 (0:00:02 remaining)  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.28s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb  
2f  
Not shown: 992 closed tcp ports (reset)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
25/tcp    filtered smtp  
80/tcp    open  http  
135/tcp   filtered msrpc  
139/tcp   filtered netbios-ssn  
445/tcp   filtered microsoft-ds  
9929/tcp  open  nping-echo  
31337/tcp open  Elite  
  
Nmap done: 1 IP address (1 host up) scanned in 15.99 seconds  
  
joshitha@joshi: ~  
$ nmap -sV scanme.nmap.org  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 15:51 +0530  
Stats: 0:00:06 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 40.00% done; ETC: 15:52 (0:00:11 remaining)  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.26s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb  
2f  
Not shown: 992 closed tcp ports (reset)  
PORT      STATE SERVICE VERSION  
22/tcp    open  tcpwrapped  
25/tcp    filtered smtp  
80/tcp    open  http  Apache httpd 2.4.7 ((Ubuntu))  
135/tcp   filtered msrpc  
139/tcp   filtered netbios-ssn  
445/tcp   filtered microsoft-ds  
9929/tcp  open  nping-echo  Nping echo  
31337/tcp open  tcpwrapped  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/.  
Nmap done: 1 IP address (1 host up) scanned in 27.14 seconds  
  
joshitha@joshi: ~  
$ nmap -O scanme.nmap.org  
Unknown argument to -O.
```



The terminal window shows the execution of an Nmap scan on scanme.nmap.org. The scan is a SYN Stealth Scan. The output indicates that the host is up and provides a list of open ports with their corresponding services. The scan took 27.14 seconds to complete.

```
joshitha@joshi: ~  
$ nmap -sV scanme.nmap.org  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 15:51 +0530  
Stats: 0:00:06 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 40.00% done; ETC: 15:52 (0:00:11 remaining)  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.26s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb  
2f  
Not shown: 992 closed tcp ports (reset)  
PORT      STATE SERVICE VERSION  
22/tcp    open  tcpwrapped  
25/tcp    filtered smtp  
80/tcp    open  http  Apache httpd 2.4.7 ((Ubuntu))  
135/tcp   filtered msrpc  
139/tcp   filtered netbios-ssn  
445/tcp   filtered microsoft-ds  
9929/tcp  open  nping-echo  Nping echo  
31337/tcp open  tcpwrapped  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/.  
Nmap done: 1 IP address (1 host up) scanned in 27.14 seconds  
  
joshitha@joshi: ~  
$ nmap -O scanme.nmap.org  
Unknown argument to -O.
```

Session Actions Edit View Help

—(joshitha@joshi)~]

—\$ nmap -A scanme.nmap.org

Starting Nmap 7.99 (<https://nmap.org>) at 2026-08-04 15:54 +0530
Stats: 0:00:02 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 17.15% done; ETC: 15:54 (0:00:10 remaining)
Stats: 0:00:28 elapsed; 0 hosts completed (1 up), 1 undergoing Traceroute
Traceroute Timing: About 32.26% done; ETC: 15:54 (0:00:00 remaining)
Stats: 0:00:29 elapsed; 0 hosts completed (1 up), 1 undergoing Traceroute
Traceroute Timing: About 64.52% done; ETC: 15:54 (0:00:01 remaining)
Stats: 0:00:41 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 96.88% done; ETC: 15:54 (0:00:00 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)

Host is up (0.13s latency).

Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:b82f

Not shown: 992 closed tcp ports (reset)

PORT	STATE	SERVICE	VERSION
22/tcp	open	ssh	OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux

x; protocol 2.0)

| ssh-hostkey:

| 1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)
| 2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)
| 256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
| 256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)

25/tcp filtered smtp

80/tcp open http Apache httpd 2.4.7 ((Ubuntu))

|_http-title: Go ahead and ScanMe!

|_http-favicon: Nmap Project

|_http-server-header: Apache/2.4.7 (Ubuntu)

135/tcp filtered msrpc

139/tcp filtered netbios-ssn

445/tcp filtered microsoft-ds

9029/tcp open nping-echo Nping echo

31337/tcp open tcpwrapped

OS fingerprint not ideal because: Host distance (23 network hops) is greater than five

No OS matches for host

Network Distance: 5 hops

Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE (using port 80/tcp)

HOP	RTT	ADDRESS
1	24.81 ms	10.72.123.21
2	...	
3	76.23 ms	255.0.0.2
4	76.57 ms	255.0.0.3
5	40.91 ms	scanme.nmap.org (45.33.32.156)

2	...	
3	76.23 ms	255.0.0.2
4	76.57 ms	255.0.0.3
5	40.91 ms	scanme.nmap.org (45.33.32.156)

OS and Service detection performed. Please report any incorrect results at <http://nmap.org/submit/>.

Nmap done: 1 IP address (1 host up) scanned in 42.61 seconds

—(joshitha@joshi)~]

—\$